Department of Nano Science Technology
Faculty of Technology
Wayamba University of Sri Lanka

Centralized Workshops and Laboratory Allocation Platform

Activity Diagram

Submitted to: Dr. U.S. Liyanarachchi
Module: NANO2142

Submitted by: Group 1

Date of Submission

05 Nov 2026

Sequence and Activity Diagrams for Critical System Flows

Implementation: C++17 / Qt Widgets / SQLite

Figure 1 documents the synchronous interaction path when a student submits a reservation request. The system executes client-side business rule validation, initiates atomic slot conflict checking through the abstract database repository, and confirms persistent reservation integrity.

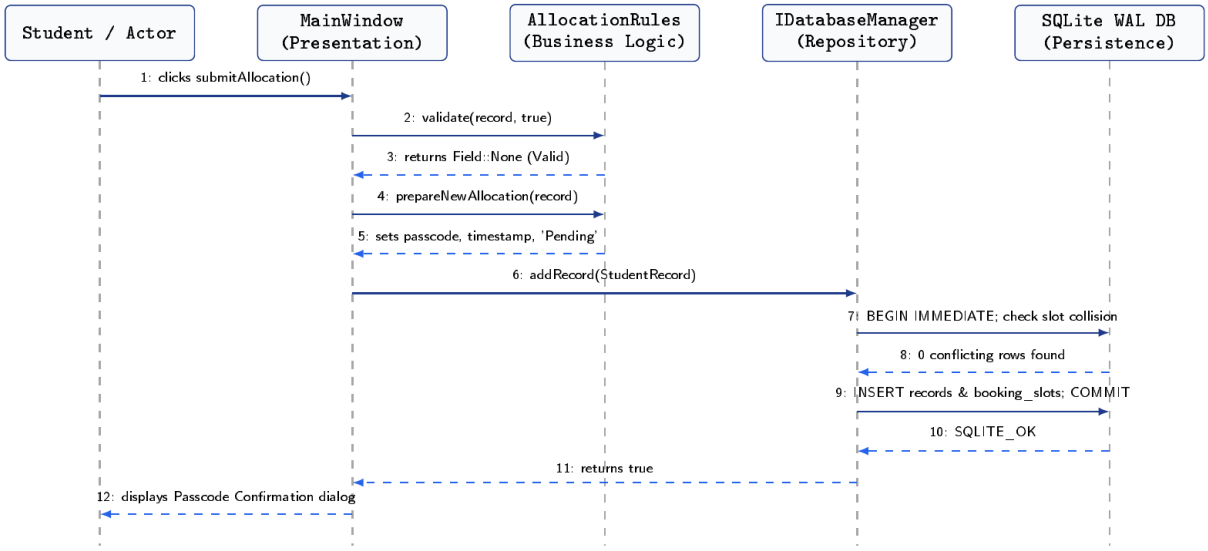


Figure 1: UML Sequence Diagram: Student Booking Validation and Atomic Persistence

Administrative Review & State Transition Lifecycle (UML Activity Diagram)

Figure 2 models the complete operational activity lifecycle for administrators reviewing submitted records, handling filtering, applying status decisions, updating atomic remarks, and triggering notification dispatch.

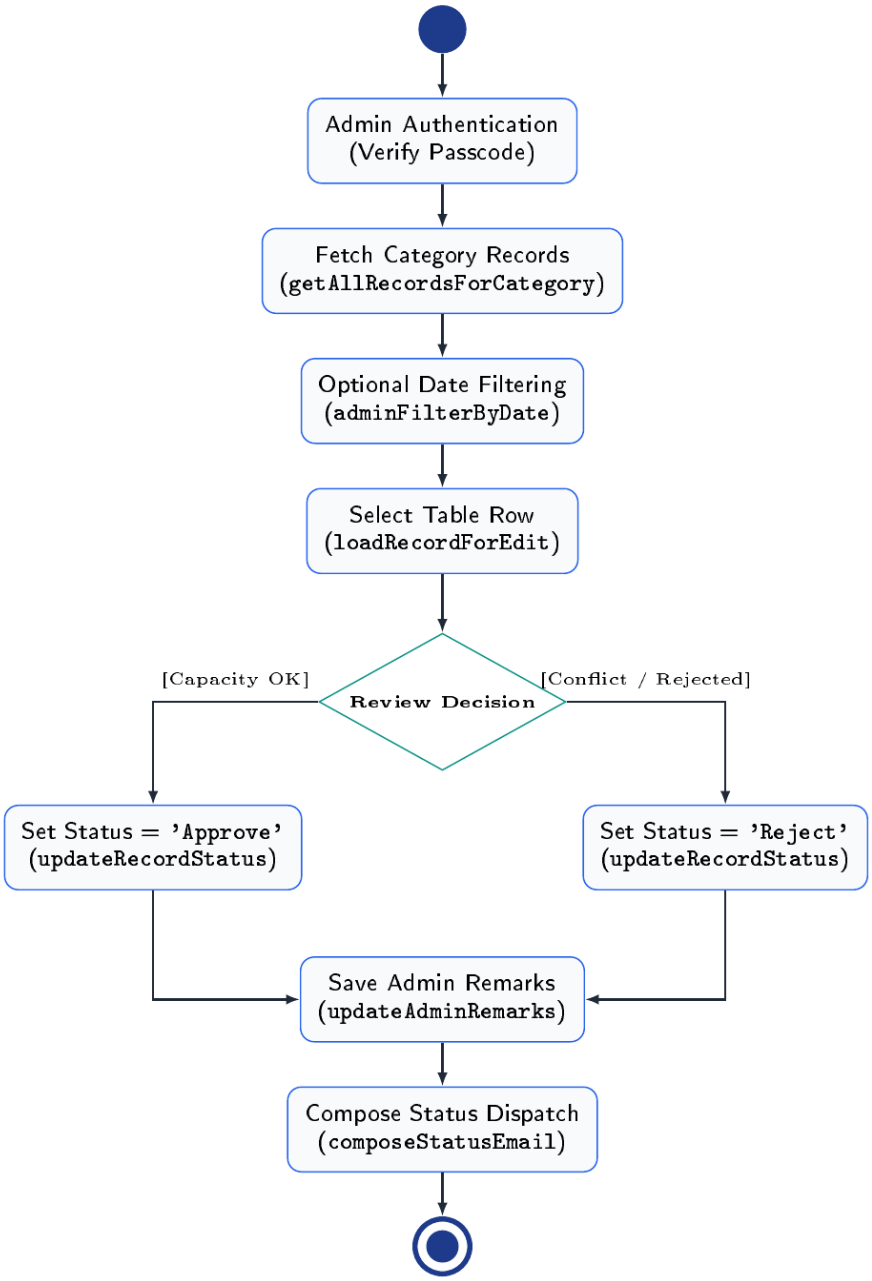


Figure 2: UML Activity Diagram: Administrative Review, Status Transition, and Notification